Compact Exhaustion Lemma

Proposition. (Compact Exhaustion Lemma)

Let $G \subset \mathbb{C}$ be an open set.

There exists a sequence $\{K_n\}$ of compact subsets of G with the following properties:

(i) $G = \bigcup_{n=1}^{\infty} K_n$ (exhaustion) $\Rightarrow G = \bigcup_{n=1}^{\infty} \operatorname{int} K_n$ open covering

(ii) for each n, one has $K_n \subset \operatorname{int} K_{n+1}$ (strictly increasing)

(iii) for each n, every component of $\mathbb{C}_\infty - K_n$ contains a component of $\mathbb{C}_\infty - G$ (no extra holes)

Compare: *Chapter VII*

1.2 Proposition. *If G is open in $\mathbb{C}$ then there is a sequence $\{K_n\}$ of compact subsets of G such that $G = \bigcup_{n=1}^{\infty} K_n$. Moreover, the sets K_n can be chosen to satisfy the following conditions:*

(a) $K_n \subset \operatorname{int} K_{n+1}$;
(b) *$K \subset G$ and K compact implies $K \subset K_n$ for some n;*
(c) *Every component of $\mathbb{C}_\infty - K_n$ contains a component of $\mathbb{C}_\infty - G$.*

Example: G = open disk $B(a; R)$

$\left\{ K_n = \overline{B}\left(a; R - \frac{R}{n}\right) \right\}$ is a compact exhaustion

Example: G = open annulus $\operatorname{ann}(a; R_1, R_2)$ $\quad (0 \le R_1 < R_2 \le \infty)$

$\left\{ K_n = \overline{\operatorname{ann}}(a; a_n, b_n) \right\}$ is a compact exhaustion

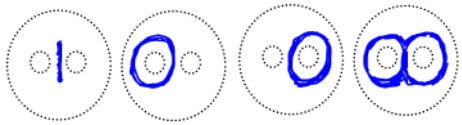
if $\{a_n\}, \{b_n\}$ are strictly monotone sequences in $(R_1, R_2) \subseteq \mathbb{R}$
with $\{a_n\} \downarrow R_1$, $\{b_n\} \uparrow R_2$, $a_n < b_n$ for each n

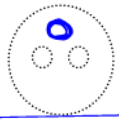
Note: Let $K \subset G$ be a compact subset.

- The unique unbounded component of $\mathbb{C}_\infty - K$ always contains a component of $\mathbb{C}_\infty - G$ (namely, the one containing ∞)
- Regard each bounded component of $\mathbb{C}_\infty - K$ as a "hole" of K
  each bounded component of $\mathbb{C}_\infty - G$ as a "hole" of G
- The condition that "every component of $\mathbb{C}_\infty - K$ contains a component of $\mathbb{C}_\infty - G$" (iii)
  means: K cannot have extra holes other than what is allowed by G

(09) for G = open unit disk $B(0;1)$

$\{ K_n = \bar{B}(0; \frac{1}{2}-\frac{1}{2n}) \cup \overline{\text{ann}}(0; \frac{1}{4}+\frac{1}{4n}, 1-\frac{1}{2n}) \}$ is a compact exhaustion but K_1, K_2 have extra holes.

(09) for G =  the K_n's should look like 

not like this 

Proof. • For each n, define

$$K_n := \bar{B}(0;n) \cap \{ z \in \mathbb{C} : d(z, \mathbb{C}_\infty - G) \geq \frac{1}{n} \},$$ compact and $\subset G$

(∵ $\inf_{w \in \mathbb{C}_\infty - G} |w-z|$)

• Clear that (i) $G = \bigcup_{n=1}^{\infty} K_n$ holds

• Also clear that (ii) holds :

$$K_n \subset B(0;n+1) \cap \{ z \in \mathbb{C} : d(z, \mathbb{C}_\infty - G) > \frac{1}{n+1} \} \subset \text{int } K_{n+1}$$

• The unique unbounded component U of $\mathbb{C}_\infty - K_n$ always contains a component of $\mathbb{C}_\infty - G$ (namely, the one containing ∞)

• Since $K_n \subset \bar{B}(0;n)$ and $\mathbb{C}_\infty - \bar{B}(0;n)$ is connected, it follows that U contains $\mathbb{C}_\infty - \bar{B}(0;n)$

• Let D be a bounded component of $\mathbb{C}_\infty - K_n$ (ie a hole of K_n)

Then D is disjoint from U and so must be $\subset \bar{B}(0;n)$

But D is disjoint from K_n, so every $z \in D$ must satisfy $d(z, \mathbb{C}_\infty - G) < \frac{1}{n}$

• Fix any $z \in D$; then there exists $w \in \mathbb{C}_\infty - G$ such that $|w-z| < \frac{1}{n}$

Then every $\zeta \in B(w; \frac{1}{n})$ satisfies $d(\zeta, \mathbb{C}_\infty - G) < \frac{1}{n}$, so $\zeta \notin K_n$

Thus $B(w; \frac{1}{n}) \subset \mathbb{C}_\infty - K_n$, and since $z \in B(w; \frac{1}{n})$, we must have

$B(w; \frac{1}{n}) \subset$ the component of $\mathbb{C}_\infty - K_n$ containing z (by connectedness) namely, D

In particular, $w \in D$

• Let D_1 be the component of $\mathbb{C}_\infty - G$ containing w

Then $D_1 \subset \mathbb{C}_\infty - K_n$, and since $w \in D_1$, we must have

$D_1 \subset$ the component of $\mathbb{C}_\infty - K_n$ containing w (by connectedness) namely, D

• Thus every bounded component D of $\mathbb{C}_\infty - K_n$ contains a component D_1 of $\mathbb{C}_\infty - G$ and (iii) holds. //

1.14 Corollary. (Runge's Approximation on open sets)

Let $G \subset \mathbb{C}$ be an open set

Let $E \subset \mathbb{C}_\infty - G$ be a subset that meets each component of $\mathbb{C}_\infty - G$.

Then for any **holomorphic function** $f: G \to \mathbb{C}$ on G

there exists a sequence of rational functions whose poles lie only in E

converging to f locally uniformly on G

compare:

1.14 Corollary. *Let G be an open subset of the plane and let E be a subset of $\mathbb{C}_\infty - G$ such that E meets every component of $\mathbb{C}_\infty - G$. Let $R(G,E)$ be the set of rational functions with poles in E and consider $R(G,E)$ as a subspace of $H(G)$. If $f \in H(G)$ then there is a sequence $\{R_n\}$ in $R(G,E)$ such that $f = \lim R_n$. That is, $R(G,E)$ is dense in $H(G)$.*

Proof. • Apply the Compact Exhaustion Lemma to get a sequence $\{K_n\}$ of compact subsets of G

such that (i) $G = \bigcup_{n=1}^{\infty} K_n$ (exhaustion)

(ii) for each n, $K_n \subset \operatorname{int} K_{n+1}$ (strictly increasing)

and (iii) for each n, every component of $\mathbb{C}_\infty - K_n$ contains a component of $\mathbb{C}_\infty - G$ (no extra holes)

- By (iii) for each n, the set $E \subset \mathbb{C}_\infty - G$ meets each (component of $\mathbb{C}_\infty - K_n$) as well
- Hence **1.7 Runge's Theorem** is applicable, and gives (with $\epsilon = \frac{1}{n}$)

a rational function $R_n(z)$ whose poles lie only in E

such that $|f(z) - R_n(z)| < \frac{1}{n}$ uniformly for all $z \in K_n$

- By (i) & (ii), for any compact subset $K \subset G$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, one has $K \subset K_n$
- Thus for all $n \geq N$, one has

$|f(z) - R_n(z)| < \frac{1}{n}$ uniformly for all $z \in K$

- This shows that the sequence $\{R_n(z)\}$ converges to f uniformly on K

Since $K \subset G$ is arbitrary compact, it follows that

$\{R_n(z)\}$ converges to f locally uniformly on G

#

Meromorphic Functions

⚠ Problematic Notion

compare: *Chapter V*

Also, to avoid the phrase "analytic except for poles" which may have already been used too frequently, we make the following standard definition.

3.3 Definition. If G is open and f is a function defined and analytic in G except for poles, then f is a *meromorphic function* on G.

Suppose that f is a meromorphic function on G and define $f: G \to \mathbb{C}_\infty$ by setting $f(z) = \infty$ whenever z is a pole of f. It easily follows that f is continuous from G into $\mathbb{C}_\infty$ (Exercise 4). This fact allows us to think of meromorphic functions as analytic functions with singularities for which we can remove the discontinuity of f, although we cannot remove the non-differentiability of f.

Riemann sphere

Qn: • With this "definition", how do/should we define $+, -, \times, \div$ on meromorphic functions ?

→ Pointwise operations are inconsistent because of ∞ : What should be the value of $0 \cdot \infty$? or $\infty - \infty$?

Eg: Compare pointwise values at $z = 0$ of

$$z^2 \cdot \frac{1}{z} \quad \text{vs} \quad z \cdot \frac{1}{z} \quad \text{vs} \quad z \cdot \frac{1}{z^2}$$

or $\frac{1}{z^2} - \frac{1}{z}$ vs $\left(\frac{1}{z} + 1\right) - \frac{1}{z}$ vs $\frac{1}{z} - \frac{1}{z}$

Exercises *Chapter V* §3

6. Let G be a region (connected) and let $H(G)$ denote the set of all analytic functions on G. (The letter "H" stands for *holomorphic*. Some authors call a differentiable function holomorphic and call functions analytic if they have a power series expansion about each point of their domain. Others reserve the term "analytic" for what many call the complete analytic function, which we will not describe here.) Show that $H(G)$ is an integral domain; that is, $H(G)$ is a commutative ring with no zero divisors. Show that $M(G)$, the meromorphic functions on G, is a field.

Qn: • How to rigorously verify associativity, commutativity, distributivity ?

How to generalize to several cplx variables ?

Definition. Let G be a Riemann surface (e.g. $G \subset \mathbb{C}$ an open set of $\mathbb{C}$ or $G = \mathbb{C}_\infty$ the Riemann sphere)

A *meromorphic function on* G is an equivalence class of data of the form

one datum:

$$\{ (U_i, g_i, h_i) : U_i \subset G \text{ open}, \quad g_i, h_i \in H(G) \text{ holomorphic functions on } G \text{ with } h_i \text{ not identically zero on any component of } G \}_{i \in I}$$

(intention: $\frac{g_i}{h_i}$; indexing)

such that • $G = \bigcup_{i \in I} U_i$ and • for any $i, j \in I$ and any $z \in U_i \cap U_j$, there exists open $V \subset U_i \cap U_j$ with $z \in V$ such that $g_i h_j = g_j h_i$ in $H(V)$ (shrink; read as $\frac{g_i}{h_i} = \frac{g_j}{h_j}$)

where $\{ (U_i, g_i, h_i) \}_{i \in I}$ and $\{ (U_j', g_j', h_j') \}_{j \in J}$ are equivalent

equiv rel:

iff$_{\text{def}}$ for any $i \in I$, $j \in J$ and $z \in U_i \cap U_j'$, there exists open $V \subset U_i \cap U_j'$ with $z \in V$ such that $g_i h_j' = g_j' h_i$ in $H(V)$ (shrinking; read as $\frac{g_i'}{h_i'} = \frac{g_j'}{h_j'}$) #

ie. a meromorphic function is locally given by $\frac{g}{h}$ fraction of holomorphic functions

Remark. Concretely, we give a meromorphic function on G ($\subseteq \mathbb{C}$) by giving:

- a closed subset $P \subset G$ with no limit points in G (set of poss. poles)
- a holomorphic function f on $G - P$ such that each $p \in P$ (isolated singularity of f) is removable or a pole

This corresponds to the data: $U_0 = G - P$, $g_0 = f$, $h_0 = 1$

and for each $p \in P$, U_p = open disk centered at p such that $U_p \subset G$ and $U_p \cap P = \{p\}$

if $m := \operatorname{ord}(f; p) \leq 0$

then $g_p(z) = (z-p)^{-m} f(z)$ extends to a function holomorphic on U_p and non vanishing at p

set $h_p(z) = (z-p)^{-m}$ holomorphic on U_p

Its equivalence class is the given meromorphic function.